Enhanced Netflix Content Recommendation Using Description Reading Time and Keyword Extraction

Ian Puszynski[**]
ipusz2@illinois.edu
puszynskian@gmail.com
University of Illinois Urbana Champaign
Champaign, Illinois, USA

Dennis Thomas
University of Illinois Urbana Champaign
dennisoct2005@gmail.com

Jackson Fleege
University of Illinois Urbana Champaign
jackson.fleege@icloud.com

Raphael Ampon
University of Illinois Urbana Champaign
amponras@gmail.com

Azaan Farooq
University of Illinois Urbana Champaign
azaanfaroog108@gmail.com

Abstract

This project aims at growing traditional content recommendation systems which are overly reliant on explicit interaction to a more sustainable enhanced model. The proposed model will uphold traditional recommendation features, while incorporating keyword extraction from description metadata as well as measuring user content engagement through time spent reading. Together these features should be representative of subthemes within genres appealing to users preferences, and more accurately capture user engagement within content browsing. Evaluating this project will require a synthetic dataset to measure the performance on past viewing experience and watchlist, comparing the new model against the traditional base model on recovering hidden dataset items from watchlists.

Keywords

Natural Language Processing, Time-Based Engagement, Keyword Extraction, Recommendation System, Content Discovery

1 Introduction

Recommendation systems are not always the most thorough when it comes to their suggestions; Users often glance over them without a second thought. Despite this, these systems remain as the most convenient option when it comes to content discovery. The issue lies with these models being overly dependent on explicit interaction, which when it comes to movies, users choose to negate this task in their time of relaxation.

A more representative statistics on a user's interest would be their time spent reading the description, as that is a frequent action, which offers a lot of implicit feedback on personal interest. Furthering that, description contains more informative metadata, which is underutilized in current filtering Methods.

In this project, we expand upon traditional recommendation systems by incorporating natural language processing to movie descriptions. Models are used to capture semantic similarity and metadata features about the movie. A hybrid model recommendation system was developed that combines TF-IDF and Word2Vec and captures the word recurrences and semantic relationships between various movies.

While timely engagement has not yet been implemented, it remains a focus of future work where it will be used as an additional measure to improve recommendation for users.

1.1 Summary

The current Netflix recommendation system estimates likelihoods that the user may like a title based on interactions with their service. Netflix also utilizes data on what members with similar tastes and preferences have watched, the time of day the user watches Netflix, preferred languages, the devices Netflix is used on, and how long the user enjoys an individual Netflix title.

Our project aims to improve this system by utilizing a natural language processor to give more engaging recommendations. Notably, Netflix tracks from titles: "their genre, categories, actors, release year, etc." Our goal is to take the descriptions of titles from Netflix and contextualize found keywords to better match shows with high user watchtime to other thematically similar shows. This will require data sets from Netflix. Recommendations will be ranked by giving each show a score based on how closely it matches a user's favorite titles. Matches in themes and overall tone will matter more than simple keyword overlap, helping surface shows that feel similar rather than just look similar on paper. Our current implementation leverages a hybrid system that combines TF-IDF and Word2Vec. Term frequency measures allow us to determine which movies feature the most similar content based on specific words such as "Cars" being used to link movies like "Fast and Furious" with "F1." Word2Vec explores semantics in descriptions and as an example can link the words "Car" and "Ferrari" together. While the current model already leverages NLP concepts and standard Netflix recommendation protocols, we hope to expand on this concept by including time-based engagement in the future. Our current model is only able to make predictions for US based content, and our pre-trained embeddings would need to be applied to foreign languages to globalize this model.

[*]Authors contributed equally to this research.

2 Description

In the work that we have done on this project, we have developed a complete recommendation system integrating data preprocessing, feature engineering, which form the bias of our hybrid retrieval model. Additionally, using AI-assistance with the permission of our project administrator, we have developed a user friendly UI to experience the results. The work on this project began with the datasets of which there are two. The first is a movie dataset containing titles and metadata stored in descriptions, genres, and countries of production. The second dataset is a synthetically generated user set which includes heldout preferences that will be used in evaluation, and their previous watch history including titles and genres. The next step is data preprocessing and normalization by changing text to lowercase, removing stopwards, punctuation and tokenization. This step aids in having data consistency and improves our overall feature extraction. In order to improve runtime, we add our post-processed data to users. Another important distinction is filtering by country which was not done in our prototype. Once data is properly cleaned it can move onto the recommendation system. The first element of this system is TF-IDF for capturing the keyword similarities across differing titles. Secondly, Word2Vec, is used to build deeper context similarity using related word semantics. The final score is calculated using a weighted scoring function of :

score = 0.30 * description-tfidf-similarity
+ 0.30 * word2vec-description-similarity
+ 0.30 * genre-tfidf-similarity
+ 0.10 * country-tfidf-similarity
+ 0.10 * genre-overlap-bonus

Finally, results are returned based on this score and returned to the user in a ranked order.

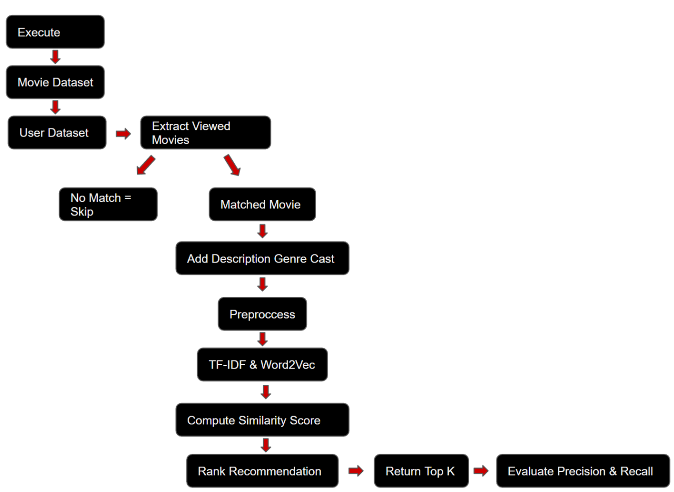

Figure 1: Workflow of the hybrid recommendation system.

3 Evaluation

In evaluating the recommender system, we used a combination of Precision and Recall for the top 25 movies per user. Results were compared against the holdout "EvaluationMoviesTheyWillLike." Precision focuses on the portion of recommendations that are relevant, and recall examines the successfulness in retrieving these recommendations.

Prototype results :
Precision = 0.0007, Recall = 0.0056
Final results :
Precision = 0.0040, Recall = 0.0250

These results demonstrate model improvement after country filtering. While still low, recommendation models like these do not always yield high scores and user satisfaction hard to classify on a specific number of a strict model as this one. Furthering this, recommendation systems like this suffer from a cold-start problem and perhaps results would be improved with even more data. Despite these lower scores, when viewing some user profiles we do see promising results with certain titles being matched with 65 percent confidence.

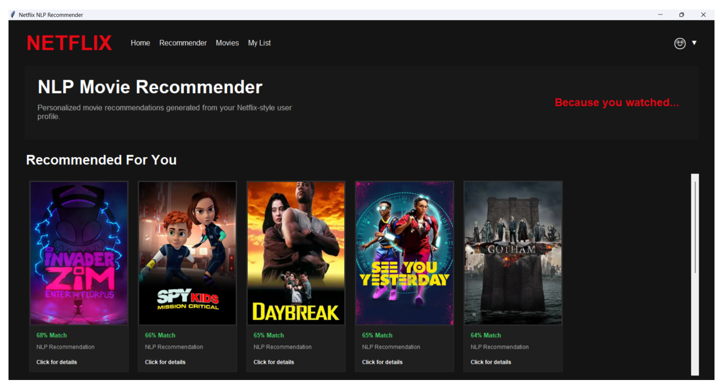

Figure 2: UI demonstrating final matches from user perspective

4 Discussion

Throughout the development of this project several important insights emerged. This project demonstrated that recommendation systems despite appearing simple can carry much difficulty due to various factors. One model cannot be used globally when handling multiple languages, and this process would require several differently trained W2V embeddings. While our model has yielded success using NLP, that is not enough and should be paired in conjunction to other recommendation systems. Our approach has been entirely content based filter, and would see benefits while being used in conjugation to collaborative filtering. While we may be able to capture deeper thematic ideas across movies, this idea should be used more as a discriminatory factor to already existing Netflix recommendations rather than being a full replacement. Finally, more developed user profiles that are accessible to current Netflix developers would be more informative than the synthetic datasets we have used. Looking forward to other opportunities to improve this work. Combining our system with a collaborative system using more developed profiles would better represent real human trends and engagement. Additionally, using time based measurements of description reading could yield valuable information on what sort of themes immediately push a user away to further discriminate on non valuable predictions. Overall, this project demonstrates the power of NLP in thematic personalization for recommendation systems.

5 Resources and Links

GitHub Repository:
https://github.com/azaan-f/netflix-nlp-recommender **Dataset 1:**
https://www.datacamp.com/datalab/datasets/dataset-r-netflix-movie-data **Dataset 2:**
https://github.com/amirtds/kaggle-netflix-tv-shows-and-movies

References

https://help.netflix.com/en/node/100639